

Function with Type Error

```
def make_line(msg:Message):
    if not isinstance(msg.body, str):
        raise ValueError("...")
1 #msg.body: str
    encode_message(msg, Utf8Encoder())
3 #msg.body: bytes
    print(msg.body + '\n')
    #TypeError: can't concat str to bytes
```

Function with Complex Type Transformation

```
def encode_message(msg: Message,
                  encoder:TextEncoder|Utf8Encoder):
    if isinstance(msg.body, str):
        msg.body = encoder.encode(msg.body)
2 #encoder: TextEncoder => msg.body: str
    #encoder: Utf8Encoder => msg.body: bytes
```

Related Classes

```
class Message:
    body: str | bytes

class TextEncoder:
    def encode(self, text: str) -> str:
        return text

class Utf8Encoder:
    def encode(self, text: str) -> bytes:
        return text.encode("utf-8")
```

Detection Results

Mypy, Pyrefly, Pyright: **No diagnostic**

PyTT: **Type error detected**